

Evaluating Uncertainty Visualizations for Geospatial Decision-Making

ANONYMOUS AUTHOR(S)

Imagine you are a consultant for the state of New Mexico. You have to recommend three counties that need financial help to support children living in poverty. Estimates in some counties are more reliable than others. Counties are only considered eligible for aid if there is a 90% chance that the poverty rate exceeds a threshold of 15%. Each map below has the same underlying data. Which counties would you pick? Which map would you prefer?

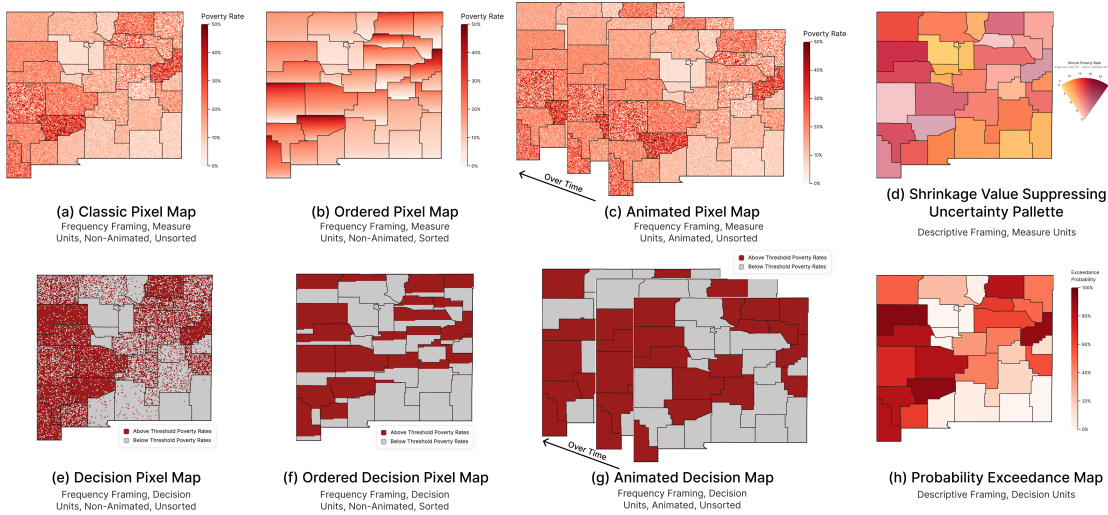


Fig. 1. Eight uncertainty visualization conditions used in the study, shown for the same county-level data and decision task.

Choropleth maps are widely used for decision-making in research and policy. Although the data they represent often contain uncertainty, it is rarely visualized because it is difficult to encode alongside estimates and may be hard for viewers to interpret. We contribute a pre-registered, crowdsourced experiment examining how people make resource-allocation decisions across eight uncertainty visualizations, chosen to represent important design dimensions for uncertainty representation. We find a tradeoff between sensitivity to evidence and bias relative to the normative decision criterion, such that visualizations presenting a more direct summary of the decision rule exhibit less variance and greater bias compared to maps that frame uncertainty as outcome frequencies. Our study shows how this tradeoff interacts with geospatial factors such as the relative size and number of areal units. Qualitative analysis of self-reported strategies characterizes how participants rely on visual heuristics, geographic context, and prior knowledge to make decisions. Our results offer guidance for responsible geospatial communication under uncertainty.

CCS Concepts: • Human-centered computing → Empirical studies in visualization.

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1 Introduction

Choropleth maps are often used to inform consequential decisions about spatially distributed phenomena, such as using geographic voting patterns for electoral planning [8], identifying areas at risk from adverse weather [26], and prioritizing interventions in areas with elevated air pollution [2]. They serve as critical tools [41] by transforming geographical statistical data into representations that researchers can use to explore spatial variation or generate hypotheses, and which policymakers rely on to observe regional disparities and allocate resources. The data behind these maps however, are often estimates derived from sample surveys, noisy measurements, and statistical models. When choropleth maps that inform consequential decisions present only point estimates without indicating their corresponding uncertainty, viewers may misinterpret the precision of the data. This can lead to overconfident or misinformed decisions, such as allocating too many resources to regions with volatile estimates, or overlooking areas where high uncertainty masks genuine need [25].

Although representing uncertainty is increasingly recognized as essential, doing so in ways that support decisions while minimizing abstract representational translations that might confuse viewers (e.g., [4, 5, 17]) remains a difficult design problem. This challenge is further exacerbated in maps, where viewers must interpret uncertainty alongside the spatial structure of the data. In this paper, we examine two design dimensions that shape the way uncertainty is transformed into geospatial decisions: the framing of uncertainty (*how uncertainty is represented*) and the unit of data representation (*what information is encoded*). While the two dimensions are often conflated, prior research has focused more on the former than the latter (e.g., [14, 16, 18, 21]). Treating these dimensions separately allows us to ask whether decision-making is shaped more by how uncertainty is framed, by what quantity is encoded, or by the interaction between the two.

The first dimension focuses on how uncertainty is framed. Some maps use a frequency-based framing, representing uncertainty through discrete visual samples of possible outcomes. For example, dither maps [22] show many plausible realizations of an uncertain estimate. The data in such maps are relatively close to the underlying distribution, but may also require viewers to visually integrate across samples of a complex display in order to make a decision. Other maps use a descriptive framing [10, 11], representing uncertainty through a summary quantity or derived visual encoding. For example, value-suppressing uncertainty palettes (VSUP) [5, 17] encode uncertainty as probability by visually attenuating less certain estimates. While such maps can reduce the amount of visual integration required from viewers, they may present a potential barrier for comprehension and need more complex statistical preprocessing within the visualization pipeline.

The second dimension focuses on what quantity the map encodes. We distinguish between two ways of encoding information – measure units and decision units. Measure units, such as in hypothetical outcome plots [14] preserve information about the underlying statistical distribution and can support multiple tasks. In contrast, decision units transform the underlying estimate into a quantity that is directly tied to a particular decision, such as the probability of exceeding a given value [20, 26]. Decision units can make a map more aligned with the decision task, but they may suppress related information and abstract away from what was measured in ways that may confuse an audience.

We study how these two dimensions influence geospatial decision making across eight visualization conditions as shown in Figure 1. Participants in our experiment play the role of consultants helping U.S. state governments allocate limited financial resources to counties with high child poverty. For each state, they review a map showing uncertain estimates of the poverty rate among children under 18 and recommend exactly three counties to receive support. They are instructed that county is eligible for support if there is at least a 90% probability that its child poverty rate exceeds 15%, providing a normative benchmark for measuring decision quality [13]. Each map communicates both estimated poverty rates per county and uncertainty in these estimates by relying on different combinations of framing and units, enabling systematic comparison across variations on these design choices.

We contribute a pre-registered¹ online experiment² on Prolific to examine how uncertainty framing and unit of representation affect decision-making with choropleth maps. We evaluate how strongly participants' recommendations for resource allocation respond to decision-relevant evidence, how systematically their selections deviate from the provided decision rule, and how these effects vary with geographic characteristics of the regions being evaluated. Rather than finding a single visualization design best suited to this task, our results show sharp trade-offs between sensitivity to evidence and bias relative to the normative benchmark. Visualizations using decision units generally promote high sensitivity to evidence but also lead to overzealous allocation of aid, a bias which is heavily exacerbated for counties with a large spatial extent and in states with large numbers of counties. Conversely, and in disagreement with prior work [14, 16, 18], visualizations relying on frequency framing to display measure units exhibit middling sensitivity but minimal bias among conditions tested. Counterintuitively, transforming the uncertainty representation to directly support a particular decision seems to backfire for supporting the targeted decision, however, such preprocessing approaches seem promising for making relevant data attributes salient to users in scenarios where there is no normative standard for decision quality. We also perform a qualitative analysis on user free-text responses about how they made decisions, how they used the given visualization, and whether they prefer choropleth maps with or without uncertainty. We find that participants use a variety of heuristics to interpret color while relying on geographical and prior knowledge to make decisions. These findings have implications for how geospatial uncertainty information can be communicated that support informed, responsible decisions.

2 Related Work

Prior work highlights the value of representing uncertainty in geospatial visualization. In his seminal work, MacEachren [23] motivates uncertainty representation as a measure of data quality that captures spatial, attribute, and temporal variation in data. Extending Bertin's original graphical variables [1], he proposes color saturation as a channel for encoding uncertainty. However, this highlights a central challenge in visualizations: the supply of perceptually distinguishable encoding channels is limited. This problem is further exacerbated in choropleths, where data values are already conveyed through color and position encodings are reserved for geographical representation so encoding uncertainty demands an additional channel that conveys meaningful variation without overwhelming the viewer. Naive solutions such as bivariate hue-saturation maps perform poorly, because the two channels are perceptually inseparable [5, 36, 38]. An effective visualization design must therefore make data values and their uncertainty separately legible while also making the uncertainty difficult to ignore.

Recent work addresses the latter challenge by integrating uncertainty more directly into the encoding. Surprise maps [4, 30] recolor regions by their Bayesian surprise relative to a reference model, thereby drawing attention to

¹https://osf.io/a836p/overview?view_only=5e3fa5e23fbc4e679496bde2aaf39869

²<https://geospatialdm.github.io/geospatial-decision-study>

unexpected values. Value-Suppressing Uncertainty Palettes [5, 17] instead allocate color non-uniformly, suppressing value resolution where uncertainty is high. Frequency-framed techniques convey the distribution over possible outcomes rather than a summary statistic: dither maps [22, 35] use visual texture while hypothetical outcome maps [14, 35] use animation over sampled draws. These approaches differ not only in what they encode but also in how they encode data.

Because our goal is to study how people make decisions, we build on a parallel line of work evaluating how uncertainty visualizations shape choices in a geospatial setting. Past work shows that the uncertainty framing and encoding change decision quality: frequency-framed and outcome-based encodings can improve calibration and choice over summary encodings in everyday decision tasks [3, 7, 15, 18]. Literature reviews of decision-making with uncertainty visualization [31] and of uncertainty representation in cartography [19, 24] highlight that effectiveness of such representations is task-dependent, yet systematic comparisons on map-based decision tasks under uncertainty, across various geographical conditions remain scarce.

Two distinctions recur across this literature focusing on a geospatial decision task: whether uncertainty is framed as *frequency* or *descriptive* summary, and whether the visualization encodes the underlying *measure* unit or the *decision* unit itself. In this study, we investigate uncertainty visualizations across these two dimensions to assess how they support resource-allocation-based tasks.

3 Uncertainty Visualization Conditions

We examine how uncertainty representations that vary in their framing and units shape decision-making with choropleth maps. Figure 1 shows the eight visualization conditions designed to instantiate combinations of these two dimensions. Within frequency-based conditions, we additionally vary layout choices such as spatial ordering and animation to probe design variations suggested by prior work [14, 29, 32, 40]. Rather than treating the eight visualizations as interchangeable alternatives, we use them to investigate fundamentally different approaches to uncertainty representation and how they change the viewer’s allocation decisions.

Classic Pixel Map (*Frequency Framing, Measure Units, Non-Animated, Unsorted*). As shown in Figure 1(a), classic pixel maps represent county-level uncertainty using a sampling-based approach. We divided each county into a grid of square pixels where each pixel represents an independent draw from that county’s poverty rate distribution. We then encode the sampled value using the Reds color scale. Thus, unlike a traditional choropleth that assigns a single color to each county, a classic pixel map displays multiple plausible poverty rates within a county. It is important to note that the spatial arrangement of pixels within a county carries no geographic meaning. Variation in pixel color represents uncertainty about the county-level poverty rate, not geographic information about sub-county regions. The map can therefore be interpreted as a static collection of hypothetical outcomes for each county. Counties with lower uncertainty appear relatively homogeneous because the sampled values are concentrated near the county estimate, while counties with greater uncertainty appear more heterogeneous because their plausible values span a wider range of the color scale.

Ordered Pixel Map (*Frequency Framing, Measure Units, Non-Animated, Sorted*). We used the same simulated poverty rate values as in the pixel map condition, but changed how those values were arranged within each county. Instead of representing values in the order in which they were sampled, we sorted the values within each county from highest to lowest before rendering the map (See Figure 1(b)). This ordering preserved the same set of plausible county level outcomes as the classic pixel map while highlighting the uncertainty as a change in gradient within the county.

Animated Pixel Map (*Frequency Framing, Measure Units, Animated, Sorted*). As in the classic pixel map condition, we divided each county into a grid of square cells, and each pixel represented one plausible poverty-rate estimate drawn

from that county's poverty estimate and standard error. However, instead of showing a single static set of sampled values, this condition displayed 50 independently sampled pixel maps as an animation (see Figure 1(c)). The animation played at 2.5 frames per second, with one frame shown every 400 ms (following prior work [14]). Within each frame, pixels were colored using the same Reds color scale. Variation across frames therefore showed repeated plausible realizations of each county's poverty rate, reinforcing the variation among pixels within a county as a representation of uncertainty in the county-level estimate.

Shrinkage Value Suppressing Palette Map (SVSUP) (*Descriptive Framing, Measure Units, Unanimated*). Shrinkage VSUPs represent uncertainty by pulling less precise estimates toward a common reference value, such that estimates with greater uncertainty are suppressed more strongly. Following Kay [17], we treat each county's poverty-rate estimate as an uncertain observation and shrink it toward a prior mean, achieving monotonic value suppression unlike the original implementation of VSUPs [5]. We set this prior mean to the global poverty rate of 14.5%, which we hold constant across all of our stimuli. Therefore, counties with larger standard errors are pulled more strongly toward the mean poverty rate, while more precise estimates remain closer to their observed values. We encode the resulting shrunk poverty rate using the YlOrRd color scale, following the use of multi-hue palettes in prior work on VSUPs [5, 17] and reduce color saturation with increasing standard error so that high uncertainty values are harder to distinguish, as shown in Figure 1(d).

Decision Pixel Map (*Frequency Framing, Decision Units, Unanimated, Unsorted*). In decision pixel maps, we represent county-level uncertainty in decision units rather than measure units. As in other pixel conditions, we divided counties into a regular grid of square cells, and each pixel represented one plausible decision outcome for that county. For each county, we first computed its exceedance probability, based on the county's poverty estimate and standard error. We then sampled each pixel from a Bernoulli distribution parameterized by the exceedance probability. Pixels with values exceeding the decision threshold were colored dark red, while pixels sampled as not exceeding the threshold were colored light gray. Counties with high poverty and low uncertainty appear more uniformly red, while counties with lower exceedance probabilities or greater uncertainty appear more mixed or more uniformly gray (see Figure 1(e)).

Ordered Decision Pixel Map (*Frequency Framing, Decision Units, Unanimated, Sorted*).

In the ordered decision pixel map condition, we used the same simulated poverty-rate values as in the decision pixel map condition, but changed how those values were arranged within each county. However, instead of displaying the simulated values in the order in which they were sampled, we sorted the pixel values within each county from exceeds threshold to does not exceed threshold. This ordering preserves the same set of plausible values as the original decision pixel map, while enabling perceptual grouping of draws that exceed vs don't exceed the threshold as shown in Figure 1(f).

Animated Decision Map (*Frequency Framing, Decision Units, Animated, Unsorted*). In this condition, we represented poverty rate and standard error for a given county using an animation on loop. We computed the probability for each county that its true poverty rate exceeded the 15% decision threshold. Each animation contained 50 frames and played at 2.5 frames per second, with one frame shown every 400 ms (as above). In each frame, each county was independently sampled from a Bernoulli distribution parameterized by its exceedance probability. Counties sampled as exceeding the threshold were colored dark red, while counties sampled as not exceeding the threshold were colored light gray. Across the animation, the frequency with which a county appeared dark red therefore approximated its probability of exceeding the 15% threshold as shown in Figure 1(g).

Probability Exceedance Map (*Descriptive Framing, Decision Units, Unanimated*). In the probability exceedance map condition, we represented county-level uncertainty as a precomputed decision-relevant probability. For each

county, we used the poverty estimate and standard error to compute the probability that the county's poverty rate exceeded the decision threshold. Unlike the frequency framing conditions, this map did not show repeated samples or variation through pixels. Instead, each county's exceedance probability was directly encoded using the Reds color scale. Counties with higher probabilities of exceeding the threshold appeared darker red, while counties with lower probabilities appeared lighter. Thus, the map processes uncertainty before rendering, transforming the estimated poverty-rate distribution into the probability of a decision-relevant outcome as shown in Figure 1(h).

4 Method

The main goal of this study was to understand how different representations of uncertainty influence geospatial decision quality and how the effectiveness of these representations depends on various geospatial characteristics.

4.1 Task and Procedure

We asked participants to take on the role of a consultant advising U.S. state governments to allocate limited financial resources to counties with acute child poverty needs. In each trial, participants viewed a county-level map of a U.S. state and were asked to recommend three counties for financial assistance. We told participants that aid should be allocated only to counties that met a predefined eligibility criterion: there had to be at least a 90% chance that the county's child poverty rate exceeded the national average of 15%. Participants therefore needed to integrate the point estimate along with the standard error to identify the three counties they believed were most likely to qualify for aid under this rule. We manipulated the datasets we showed in each map so that there were always exactly three "correct" answers based on the endowed decision rule, so we can describe user responses in reference to a clear normative standard of decision quality.

Payment. All participants were awarded a guaranteed reward of \$5. Additionally, participants could earn a \$0.10 bonus for every correct county they chose and lose a \$0.05 bonus for every incorrect county they selected. Participants could thus earn a minimum reward of \$5 and a maximum reward of \$9.8. These values guaranteed a pay higher than the state minimum wage for all participants, a bonus for those who carefully made their decisions, and no extra pay for those who made random selections.

User Familiarity, Strategies, and Uncertainty Preferences. We asked participants which states they were familiar with or had lived in, so we could use this as a moderating variable in our analysis of decision quality. To learn more about how participants chose which counties to allocate financial resources to, at the end of the experiment, we showed them an example map of the condition they were assigned and asked *"You have seen many example maps like the one above. How did you use the map to recommend counties that should receive financial assistance? Please describe your thought process in as much detail as possible."*

We were also interested in learning whether participants preferred maps with uncertainty for this task vs choropleth maps that they may be more familiar with. It is important to note that the choropleths had no uncertainty information, which was necessary to do this task. We wanted to know if participants could accept the complexity of uncertainty information and recognize its necessity to the task while making their choice. Participants used a button to make their choice. We also asked them, *"Above, you can see two maps based on the same underlying data. The map on the right shows estimated poverty rates only. The map on the left (the one you just used) shows estimated poverty rates along with uncertainty information. Which of the two maps would you prefer to use for making decisions like the one in this experiment? Please explain your choice."*

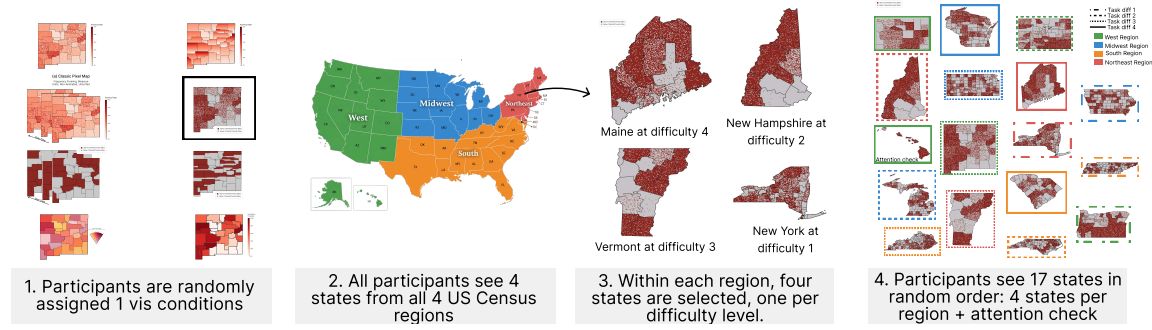


Fig. 2. Study assignment and trial-sampling procedure. Participants were randomly assigned to one visualization condition and completed 16 state trials spanning four Census regions and four difficulty levels, plus one attention check.

4.2 Experimental Design

We deployed the study interface using ReVISit [6], a web browser based framework to conduct user studies. As shown in Figure 2, each participant completed 16 experimental trials, viewing four states from each of the four U.S. Census regions. Within each region, the four states were assigned to the four task-difficulty levels (1–4), such that each participant encountered one state at each difficulty level per region. State assignment, difficulty level, and visualization condition were counterbalanced across participants. Participants also completed one attention-check trial using Hawaii, for a total of 17 decision trials. Because participants selected exactly three counties per trial, each participant selected 51 counties in total.

We excluded Delaware from the experimental trial pool because it contains only three counties. Hawaii was reserved for the attention check, and Alaska was used only for a practice trial to familiarize participants with the task interface before the experiment began.

4.3 Data and Generating Stimuli

To balance ecological validity with experimental control, we generated semi-synthetic county-level stimuli from real ACS poverty estimates and margins of error. Our data generating process preserved each state’s geography, county ranking, and uncertainty structure while systematically manipulating the strength and distribution of decision-relevant evidence across four difficulty levels.

Data. To improve the ecological validity of our task, we used real-world estimates from the 2022 American Community Survey (ACS) 5-year data. ACS tables report margins of error, typically at the 90% confidence level, for published estimates. This made the ACS well suited for our experiment, in which participants were asked to reason about both estimated poverty rates and the uncertainty associated with those estimates. We used county-level estimates of poverty among people under 18 years of age, together with the corresponding margins of error as raw data.

Task Difficulty. We operationalized four levels of task difficulty by manipulating the distribution of evidence in the decision-relevant signal space, where evidence is defined as the probability that a county’s poverty rate exceeds the 15% threshold (see Figure 3). Each difficulty level was jointly operationalized by four parameters:

- (1) **Qualifying-county evidence center:** the central exceedance probability of the three ground-truth counties. Lower values place the qualifying counties closer to the 0.90 eligibility criterion.

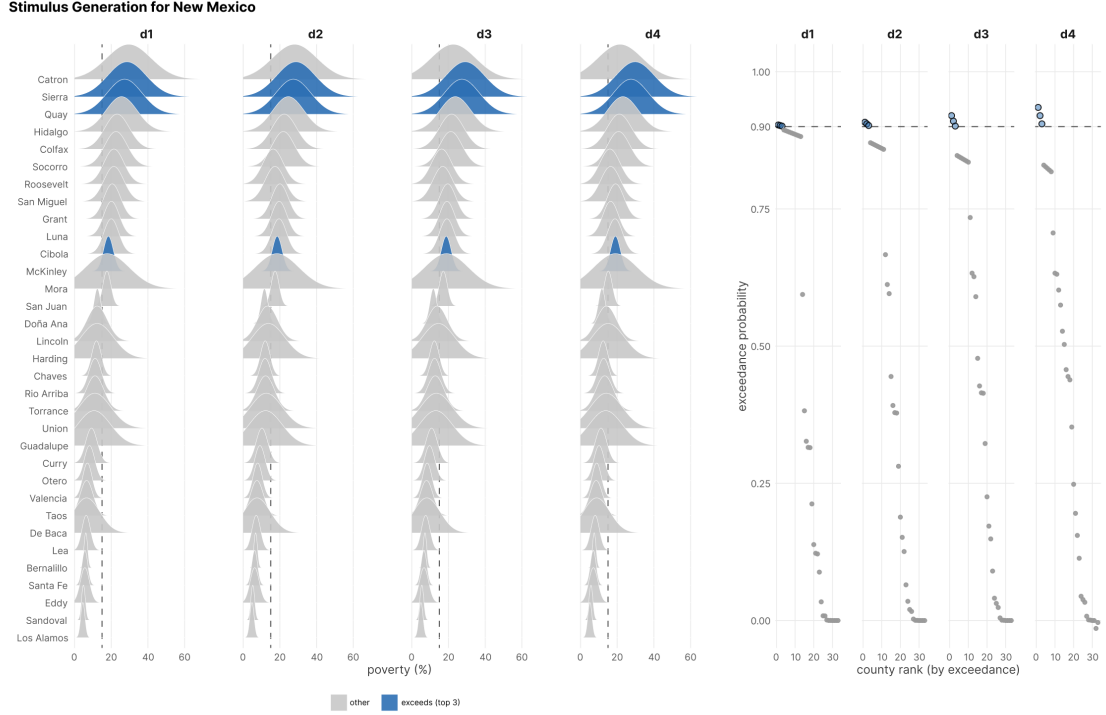


Fig. 3. Semi-synthetic stimulus generation for New Mexico across four difficulty levels. County poverty distributions were adjusted so that exactly three counties exceeded the 0.90 eligibility criterion at each difficulty level, while preserving county-level uncertainty and rank structure. Left: county poverty-rate distributions relative to the 15% threshold, with qualifying counties highlighted. Right: corresponding exceedance probabilities ranked by county, with the dashed line marking the 0.90 decision criterion.

- (2) **Qualifying-county spread:** the separation among the exceedance probabilities of the three ground-truth counties. For a center c and spread s , their target exceedance probabilities were $c + s$, c , and $c - s$. Smaller values of spread place qualifying counties closer to each other.
- (3) **Qualifying-decoy distance:** the difference in log-odds of mean exceedance probability between the qualifying counties and the plausible decoys. Smaller distances place the decoys closer to the qualifying counties, making them harder to distinguish.
- (4) **Decoy fraction:** the fraction of counties assigned to the near-threshold decoy set. Larger fractions introduce more plausible alternatives to the three ground-truth counties.

Since task difficulty is a multidimensional construct, we needed to jointly manipulate the four parameters to determine the discriminability of the qualifying counties. We also needed a way to normalize the process given the variability in our data. For instance, we used fraction instead of a fixed value to determine the number of decoy since the number of counties in our data ranged between 5 (in Rhode Island) to 254 (in Texas). Similarly, standard error, which is necessary to calculate exceedance probability for qualifying as well as decoy counties ranged from (0.4, 42.25). We also had a fixed mean of 14.5% for all variations of our data for experimental control. Across the four difficulty levels where difficulty goes

from most difficult to least, the qualifying-county evidence center increased as $\{0.902, 0.905, 0.910, 0.920\}$, the qualifying-county spread increased as $\{0.001, 0.003, 0.010, 0.015\}$, the qualifying-decoy distance increased as $\{0.15, 0.40, 0.65, 0.90\}$, and the decoy fraction decreased as $\{0.30, 0.25, 0.20, 0.15\}$. Thus, the more difficult stimuli placed the three qualifying counties closer to the eligibility criterion and closer to a larger set of plausible alternatives.

We constructed four semi-synthetic county-level datasets for each state using observed American Community Survey (ACS) child-poverty estimates and margins of error. Our generation procedure operated first in the decision-relevant signal space: the probability that a county's poverty rate exceeds the eligibility threshold. We then back-transformed these probabilities into county-level poverty means that could be visualized.

For each county i , we converted the ACS 90% margin of error to a standard error,

$$SE_i = \frac{MOE_i/100}{1.645},$$

where 1.645 is the z-score corresponding to a 90% confidence interval. We used the observed ACS poverty estimates to rank counties within each state and used this ranking to determine which counties occupied the qualifying, decoy, and remaining positions in the constructed stimulus. Therefore, even though the poverty estimates we use in our study are different from the original ACS dataset, the rank order and the standard error of counties are preserved. This allows us to have experimental controls while ensuring that the relative values of poverty rate that participants observe across counties in a state resemble real world values and enabling them to use prior knowledge when useful (See Section 5 to learn about familiarity of states).

We modeled the poverty estimate for county i as

$$Y_i \sim \mathcal{N}(\mu_i, SE_i),$$

and defined its task-relevant signal as its exceedance probability

$$p_i = P(Y_i > 0.15).$$

Rather than directly manipulating the poverty mean μ_i , we first specified a desired value of p_i and then solved for the corresponding mean:

$$\mu_i = 0.15 + \Phi^{-1}(p_i)SE_i.$$

This transformation guarantees that

$$P(Y_i > 0.15) = p_i.$$

Consequently, the experimental structure could be controlled directly in terms of the evidence participants needed to use for the task.

For each state and difficulty level, we assigned exactly three counties to be the ground-truth qualifying counties. Given a qualifying-county evidence center c and spread s , their target exceedance probabilities were

$$p_{\text{qualifying}} = \{c + s, c, c - s\},$$

with all three constrained to exceed the 0.90 eligibility criterion.

We next constructed a set of near-threshold *decoy* counties. Let $\bar{p}_{\text{qualifying}}$ denote the mean exceedance probability of the three qualifying counties. For a target qualifying-decoy distance D , we defined the central decoy exceedance probability as

$$p_{\text{decoy}}^* = \text{logit}^{-1}(\text{logit}(\bar{p}_{\text{qualifying}}) - D).$$

Equivalently, the manipulation targets the log-odds separation

$$D = \text{logit}(\bar{p}_{\text{qualifying}}) - \text{logit}(\bar{p}_{\text{decoy}}).$$

Individual decoy probabilities were distributed around this center and constrained to lie between 0.80 and 0.899, ensuring that they were plausible alternatives while remaining below the eligibility criterion. The number of decoys was determined by the difficulty-specific decoy fraction.

The remaining counties formed a lower-evidence tail, with exceedance probabilities constrained to be at most 0.78. We shifted this tail distribution so that the unweighted mean of the generated county poverty means was fixed at 14.5% for every state $\times$ difficulty combination. Generated county poverty means were additionally constrained to lie between 4% and 50%.

Finally, we back-transformed each constructed exceedance probability to its corresponding poverty mean using the county-specific standard error from the ACS data. We validated each generated state $\times$ difficulty dataset to ensure that exactly three counties satisfied

$$P(Y_i > 0.15) \geq 0.90,$$

that the mean generated county poverty rate was 14.5%, and that all generated poverty estimates fell within the specified bounds. This procedure preserves realistic state geography and county-specific uncertainty while allowing us to experimentally control the strength and distribution of evidence relevant to the resource-allocation decision.

4.4 Modeling

We wanted to measure how uncertainty representations and geospatial characteristics influence both participants' sensitivity in distinguishing between counties and their systematic bias in county selection. To measure sensitivity, we fit a Bayesian multilevel logistic regression to county-selection decisions and derive just-noticeable differences (JNDs), which describe the scale of the selection curve and thus how much evidence is needed to produce a detectable change in selection behavior. To measure bias, we derive points of subjective equality (PSEs), which describe the location of the selection curve and indicate the level of evidence at which counties are equally likely to be selected according to our decision criterion.

We relied on the principle of model expansion to incrementally build up a model that accounts for the maximum possible amount of structure that we can fit in our data. We followed a procedure similar to prior work (cf. [9, 15]), enabling us to draw robust inferences about conditional expectations for our main performance measures (JND and PSE). See our preregistration³ for details about this procedure, and see Supplemental Material for our implementation.

³https://osf.io/a836p/overview?view_only=5e3fa5e23fbc4e679496bde2aaf39869

Logistic Regression. We use the following model to make inferences about county selection decision.

$$\begin{aligned}
 \text{Line 1: } & y_{ptc} \sim \text{Bernoulli}(p_{ptc}) \\
 \text{Line 2: } & \text{logit}(p_{ptc}) = \text{evidence} \times \text{condition} \times \left(n \text{ counties} + SE \right. \\
 & \quad + \text{area} + \text{size ratio} + \text{neighbor poverty ratio} + \text{familiarity} \\
 & \quad + \text{region} + \text{border} + \text{compactness} + \text{difficulty} \\
 & \quad \left. + \text{Moran's } I + \text{island} \right) \\
 \text{Line 3: } & + \text{evidence} \times \left(\text{neighbor count} + \text{urban} + \text{reservation} \right) \\
 \text{Line 4: } & + \text{Moran's } I \times n \text{ counties} \\
 \text{Line 5: } & + \text{neighbor poverty ratio} \times \text{neighbor count} \\
 \text{Line 6: } & + \text{size ratio} \times \text{neighbor count} \\
 \text{Line 7: } & + \text{size ratio} \times \text{neighbor poverty ratio} \\
 \text{Line 8: } & + \text{area} \times \text{compactness} \\
 \text{Line 9: } & + \text{familiarity} \times \left(\text{urban} + \text{reservation} + \text{area} \right. \\
 & \quad \left. + \text{border} + \text{compactness} + n \text{ counties} \right) \\
 \text{Line 10: } & + (1 \mid \text{participant}) + (1 \mid \text{state}) \tag{1}
 \end{aligned}$$

where $\times$ denotes an interaction together with all constituent lower-order terms, and $+$ denotes a main effect only. Continuous predictors are logged where skewed and then standardized. Binary predictors are centered to $-0.5/+0.5$. *Condition* is coded with sum contrasts against a retained global intercept. The unit of observation, indexed ptc , is one county c on one trial t for one participant p as selection is recorded for every county on every map, not only for the three a participant chooses.

Line 1: *selection modeled as a Bernoulli outcome.* The dependent variable y_{ptc} is 1 if participant p recommended county c for aid on trial t and 0 otherwise. We model it using a Bernoulli distribution with probability p_{ptc} and a logit link, so every coefficient below is a change in the log odds that a county is selected.

Line 2: *evidence, visualization condition, and predictors.* *Evidence* (s_c) is the log odds that a county's true poverty rate exceeds the 15% decision threshold and centered at the grand mean of the analysis sample. Larger values mean stronger evidence that the county qualifies. Unlike the other continuous predictors, evidence is centered but deliberately *not* standardized, so that its coefficient stays in log-odds-per-log-odds units: a slope of 1 describes a participant whose log odds of selecting a county track the log odds that the county qualifies one for one. This is the scale on which the just-noticeable difference (JND) is defined, and keeping it fixed makes the JND comparable across conditions.

Condition is the visualization type, manipulated between subjects: classic pixel maps, ordered pixel maps, animated pixel maps, shrinkage value suppressing uncertainty palette maps, decision pixel maps, ordered decision pixel maps, animated decision maps, and exceedance probability maps. The remaining twelve predictors interact with evidence and condition. The twelve predictors are:

- *n counties* - the number of counties in the state on that trial, logged and standardized. Note that this predictor also carries information about our experimental design: since we ask participants to choose exactly three out of *n* counties, the baseline selection rate falls as *n* rises. The range for *n* is [5, 254].
- *SE* - the standard error of the county's displayed poverty estimate (*moe*/1.645), ranging from (0.40, 42.24), logged and standardized. This is the magnitude of the uncertainty the visualization had to convey for that county.
- *area* - the pixel area the county occupies as rendered in the experiment, logged and standardized. Higher values mean a physically larger, more visually salient county.
- *size ratio* - rendered pixel area divided by the mean rendered area of the county's neighbors, logged and standardized. This separates being large in absolute terms from standing out as large relative to a county's neighbors.
- *neighbor poverty ratio* - the log of the county's poverty estimate divided by the mean poverty estimate of its neighbors, standardized.
- *familiarity* - whether the participant named this state as one they are familiar with, centered to ± 0.5 .
- *region* - the Census region of the state (Northeast, South, Midwest, West). Each participant sees four states from each region at a unique difficulty level.
- *border* - whether the county shares more than a meter of edge with the state's outer boundary, centered to ± 0.5 .
- *compactness* - the Polsby - Popper score, $4\pi \text{ Area}/\text{Perimeter}^2$, bounded on (0, 1], standardized without logging since it is bounded and not strongly skewed. Higher values mean a county has a more circular shape.
- *difficulty* - the task difficulty level of the trial, an ordered factor from most to least difficult. See Section 4.3 to learn more about how we operationalize task difficulty.
- *Moran's I* - the global Moran's I of the poverty estimate for that state $\times$ difficulty combination, computed with row-standardized Queen weights and standardized. It measures how spatially clustered poverty is for a given state. A state where poverty is strongly clustered is one where most counties resemble their neighbors. This term lets us ask how that global structure shapes selection.
- *island* - whether the county has no contiguous neighbors, centered to ± 0.5 . Islands are very rare in our data, so these estimates lean heavily on the prior and we interpret them cautiously.

Line 3: *predictors that may moderate evidence sensitivity but are not expected to interact with condition.* *Neighbor count:* the number of Queen-contiguous neighbors, standardized, *urban:* whether the county falls in any Census 2020 urban area, and *reservation:* whether the county overlaps an American Indian, Alaska Native, or Native Hawaiian area by at least 1% of its land.

Lines 4-8: *Interactions among predictors.* We included these five pairs to study how two features of a map combine to guide decision-making. *Moran's I* $\times$ *n counties* (Line 4) allows the effect of spatial clustering to depend on how many counties are on the map, since clustering in a 15-county state and a 250-county state lead to different visual stimuli. Lines 5-7 tie the two relative measures to the neighborhood they are computed over: a ratio to one's neighbors means something different when a county has two neighbors vs when it has ten (Lines 5 and 6). Line 7 asks whether a county that is both larger and poorer than its neighbors is selected more than the two effects separately would suggest. Line 8 allows the effect of rendered area to depend on shape, to study if a large but contorted county selected as often as a large compact one.

Line 9: familiarity and visual salience. We want to evaluate if prior knowledge of a state's geography and economy gives a participant something to rely on besides the visual stimuli.

Line 10: group-level effects. Participants differ in baseline selectivity, and states differ in their geography and in features we have not measured. We therefore include a varying intercept for participant and a varying intercept for state.

4.5 Derived Measures

Design Anchored JND. For each visualization condition, we estimate the JND from the logistic model after marginalizing over the other predictors. We define the JND relative to the design-relevant baseline selection probability $3/n_{\text{counties}}$ (i.e., guess rate), corresponding to selecting three counties from a state. Specifically, $[(\text{logit}(3/n_{\text{counties}}) + \log(2)) - \text{logit}(3/n_{\text{counties}})] / b$. The JND therefore represents the increase in evidence required to double the odds of selecting a county. This is our primary measure of discriminability, and the comparisons that bear on our main hypothesis are two contrasts over conditions: the six frequency-based conditions against the two descriptive conditions, and within the frequency-based set, the three outcome-unit pixel maps against the three thresholded variants. We will also compare ordered against unordered pixel maps, and ordered against unordered decision pixel maps.

Design Anchored PSE. The canonical PSE for each visualization condition, marginalizing over the remaining predictors, is the level of evidence s at which the predicted probability of selecting a county is 0.5. Because evidence is centered at its grand mean $\bar{s}$, this is calculated as

$$s_{\text{PSE}} = \bar{s} - \frac{a}{b},$$

where a is the condition-specific intercept and b is the condition-specific evidence slope, and is then back-transformed to the exceedance-probability scale. It represents the level of evidence at which a participant is equally likely to recommend or not recommend a county, and thus provides a measure of bias in their decision criterion relative to the instructed 90% threshold. In our experimental design, participants can recommend only three counties from a state containing many alternatives. Consequently, the baseline probability that any one county is selected is substantially below 0.5 (i.e., below the canonical PSE for a 2AFC task). If the fitted selection curve does not reach a predicted selection probability of 0.5 within the observed range of evidence, the canonical PSE lies outside the empirical support of the data and requires extrapolation. We therefore report a design-anchored PSE: the level of evidence at which the predicted selection probability equals $3/n_{\text{counties}}$, calculated as

$$s_{\text{PSE}}^{\text{design}} = \bar{s} + \frac{\text{logit}(3/n_{\text{counties}}) - a}{b}$$

This quantity anchors the decision criterion to the selection rate implied by the task and is the version of the bias measure we will interpret substantively.

4.6 Participants

We recruited users through Prolific. We aimed to recruit 800 participants, 100 per uncertainty representation condition. We determined this target sample size based on a heuristic power analysis using our pilot data, specifically the number of participants we expected would be needed to reliably detect the effect of sorting vs not sorting hypothetical measure units in a pixel map. We chose this comparison as the basis for our power analysis because we believe it reflects the smallest conceptually important between-subjects difference we want the experiment to inform on. Accordingly, we

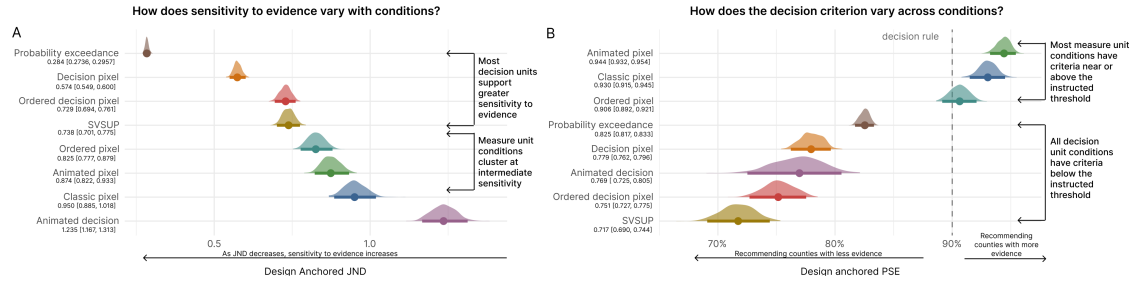


Fig. 4. Sensitivity to evidence and decision criteria across visualization conditions.

anticipated being able to detect most moderating effects of geospatial factors on decision quality, since these were measured across many trials within-subjects. We erred on the side of over-powering the study, so our results could provide unambiguous practical guidance to cartographers and GIS specialists. However, when we initiated the model expansion process with 20% of the data—searching for a well-fitting model that captured maximal structure without prematurely probing statistical inferences—we realized that that 5x as much data would likely be overkill and would make the model fit objects unwieldy. Accordingly, we decided to balance our sample across visualization conditions and stop data collection early, thus deviating from our preregistration. Our prospective power analysis suggested that our realized final sample size of 82 participants per condition would have 70% power to detect the target effect of sorting.

After excluding the attention check, each participant contributed 16 trials and, because selection was recorded for every county shown, we received 730 to 1,242 county-level observations per participant, giving 640,765 observations in total. We required participants to reside in the United States, which was necessary for the familiarity hypothesis and for meaningful interpretation of state-level geography, and to be fluent in English. Participants who took part in the pilot study were excluded from the main study. After verifying their responses to the attention check, such that at least 2/3 counties for HI were correct, we collected data until we had 656 participants. All participants were paid regardless of exclusions.

4.7 Qualitative Analysis

We conducted qualitative analysis based on the two questions we asked participants about their strategy to choose counties for resource allocation and their preference between their assigned uncertainty map and a choropleth. We used an open coding approach to identify the common strategies participants used to recommend counties, whether they preferred uncertainty visualizations or choropleths and why.

5 Results

We present our results in three parts. In Section 5.1, we describe how participants selected counties based on design anchored PSE and design anchored JND for each visualization condition. Next, in Section 5.2, we expand on how different geospatial properties influenced the decision of which counties should be recommended for financial support. Lastly, we present our analysis of the strategies and preferences participants described while doing the study through their survey responses in Section 5.3. While we focus on comparisons most relevant to our main findings for brevity

here, we report additional analyses in the Supplemental Materials. Omitted reporting presented in the Supplement includes null results for preregistered comparisons on the following geospatial factors: familiarity, urban, reservation, and neighbor count. Results reported in figures and 95% containment intervals throughout this section represent marginal comparisons of push-forward conditional expectations derived from the Bayesian regression model described in Section 4.4, integrated over observed trials within each condition we tested in our experiment.

5.1 How does framing of uncertainty and unit of representation affect geospatial decision-making?

While visualization conditions that encode the decision units can support greater sensitivity to evidence, their decision criterion is systematically below the instructed threshold. As shown in Figure 4, probability exceedance, decision pixel, and ordered decision pixel maps had design-anchored JNDs of 0.284, 0.574, and 0.729, respectively, indicating greater sensitivity to evidence than most other conditions. Recall that design anchored JND represents the increase in evidence required to double the odds of selecting a county. At the same time, their corresponding PSEs were 0.825, 0.779, and 0.751, all substantially below the instructed 0.90 decision criterion. This suggests that directly representing information in units that correspond to the decision rule can make participants more responsive to changes in decision-relevant evidence, without necessarily bringing their decision criterion closer to the instructed rule.

This pattern is somewhat counterintuitive. As decision units based visualizations directly encode information relative to the task’s decision rule, they reduce the representational translation required from the viewer. One might therefore expect greater alignment with the instructed 0.90 criterion. Instead, although participants were relatively sensitive to changes in evidence, they tended to select counties at lower levels of evidence than warranted by the rule. Thus, greater sensitivity to evidence does not imply better alignment with the instructed decision criterion. While our quantitative analysis cannot determine why this pattern occurs, we return to participants’ reported decision strategies in the qualitative analysis to better understand how they interpreted these representations.

This sensitivity pattern did not extend to all decision-unit representations. Animated decision maps had the largest design-anchored JND, 1.235 [1.167, 1.313], indicating substantially lower sensitivity to evidence, while their PSE also remained well below the instructed criterion at 0.769 [0.725, 0.805]. Animated decision maps therefore combined low sensitivity with substantial downward bias relative to the decision rule.

Frequency based maps encoding measure variables, on the other hand, show a contrasting pattern, with lower sensitivity to evidence but decision criteria that were closer to the instructed threshold. Classic pixel maps, animated pixel maps, and ordered pixel maps produce PSEs at or above the decision rule, but provide less sensitivity to evidence. Animated pixel maps have the highest PSE at 0.944, followed by classic pixel maps at 0.930. Only ordered pixel maps are approximately aligned with the rule at 0.906, with the 95% interval spanning 0.90. At the same time, these three conditions show relatively similar, albeit lower sensitivity to evidence. Only ordered pixel maps separate reliably from classic pixel maps in marginal contrasts of conditional JND estimates, while animated pixel maps separate from neither static counterpart.

These findings reflect a bias-variance trade-off whereby a simplifying representation tends to promote higher sensitivity. Specifically, we observe that visual representations which rely on concomitant data transformations [28, 39] that reduce the entropy of measured data — such as decision unit visualizations and SVSUPS — tend to both increase bias and reduce variance in decision-making. We posit that this may be a *canonical effect of entropy-reducing data transformations*, and we will use this shorthand to describe this pattern throughout the results section hereafter. Instead of identifying a single visualization as the best choice for this task, our results suggest that different representations

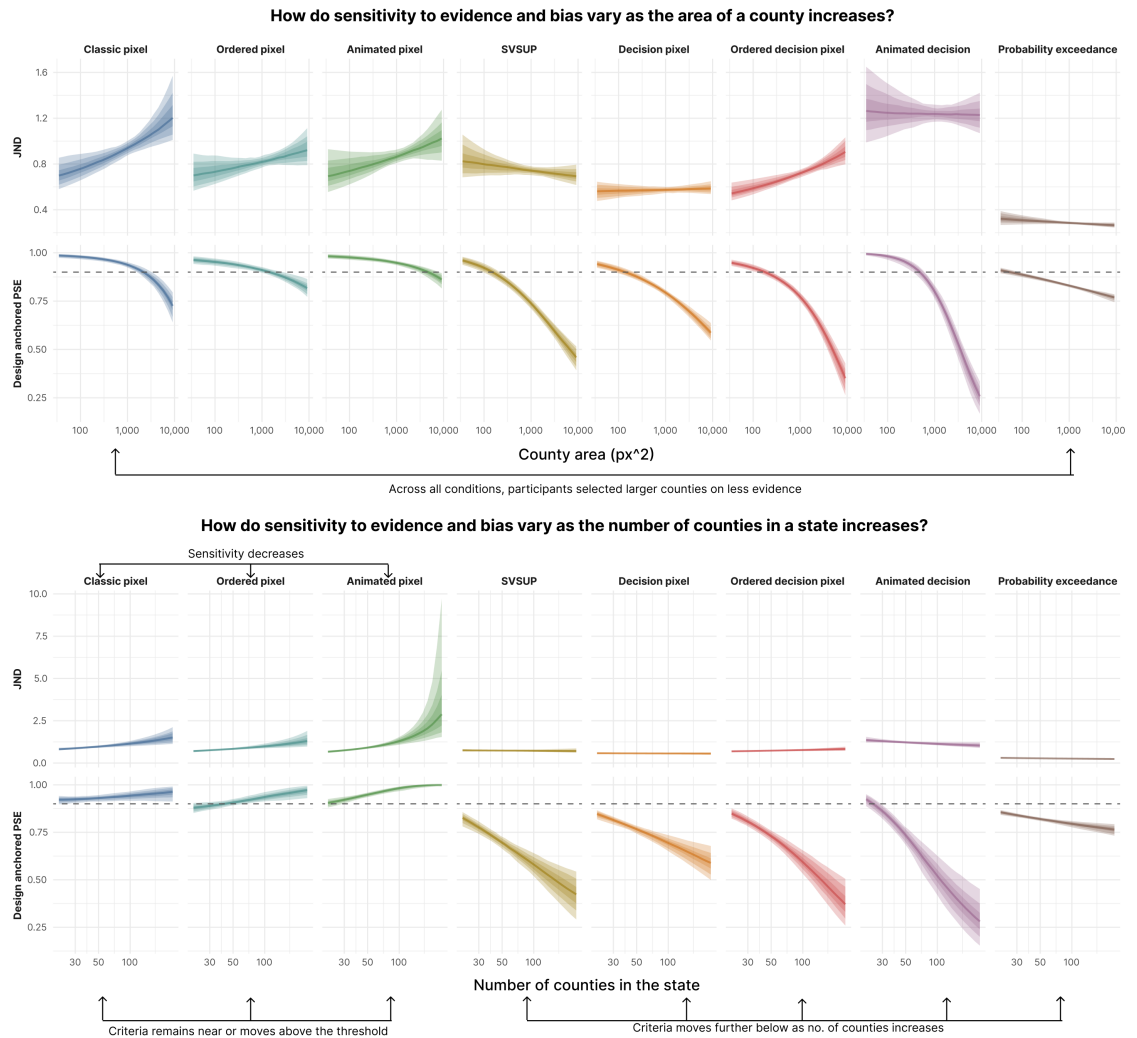


Fig. 5. Effects of county area and map size on sensitivity and decision criterion across visualization conditions. Top: Estimated JND and design-anchored decision criterion as county area increases. Bottom: Estimated JND and criterion as the number of counties in a state increases.

supported different aspects of decision-making that varied with the geographic characteristics of the regions being visualized, as we show in the next section.

5.2 How does geography influence decision-making?

Across all conditions, participants selected larger counties on less evidence. As shown In Figure 5, of all the predictors we examined, county area was the one where the designed anchored PSE decreased across all eight conditions. This means that a county occupying larger space on the map was selected at a lower exceedance probability than a

smaller county. Interestingly, the two animated conditions lie at the ends here: animated pixel maps showed the smallest shift (PSE -0.121) as county area increases while animated decision maps showed the largest (PSE -0.735). This might be in part as changes in animated decision units are harder to track than changes in animated measure units. We also found that changes in the design anchored JND were confined to two conditions (classic pixel and ordered decision pixel), so area acted almost entirely on the selection criterion rather than on sensitivity to evidence.

The number of counties in a state was also associated with how participants made decisions. We found that as the number of counties within a state increased, participants' sensitivity declined for classic, ordered, and animated pixel maps i.e., they required a larger increase in evidence to produce the same change in a county's odds of being selected when choosing among more counties. Animated decision and probability exceedance maps showed modest decrease in sensitivity as number of counties increase, while the remaining conditions changed comparatively little. The number of counties in a state also shifted participants' decision criteria, but in different directions across conditions as shown in Figure 5. As the number of counties increased, the criteria for classic, ordered, and animated pixel maps remained near or moved above the instructed 0.90 threshold, indicating that participants required at least as much evidence as prescribed by the decision rule. In contrast, the criteria for SVSUP and the decision unit conditions moved farther below 0.90 as the number of counties increased.

County selection was also associated with a county's size relative to its neighbors. Across all visualization conditions, design anchored JND increased as a county became larger relative to neighboring counties, indicating reduced sensitivity to evidence. In other words, when a county occupied more visual area than its neighbors, participants required a larger increase in evidence to produce the same change in its odds of being selected as shown in Figure 7. Relative county size also affected decision bias, but the pattern differed across visualization designs. Visualizations with frequency framing that encoded measure units remained comparatively close to the instructed 0.90 decision criterion as relative county size changed, indicating relatively little bias. In contrast, all conditions that encoded decision units had criteria farther below 0.90, illustrating the canonical effect of entropy-reducing data transformations. **Additionally, when a county's poverty rate increased relative to its neighbors, participants became more sensitive to evidence across all conditions, while decision criteria generally shifted closer to the instructed rule.** The magnitude of this improvement varied across conditions, with probability exceedance maps changing the least. This pattern may partly reflect local visual contrast but our qualitative analysis suggests participants also deliberately used neighboring counties as a basis for comparison. Decision criteria also became more similar across most conditions as relative poverty increased. Frequency based maps encoding measure variables, whose criteria were initially near or above the 0.90 criterion, shifted downward, whereas SVSUP and most decision-unit conditions shifted upward from criteria below the rule as shown in Figure 7. Animated decision maps were an exception, with their criterion shifting further below the instructed threshold. The influence of local relative poverty in allocation behavior was also reflected in participants' reported strategies, where comparisons with neighboring counties emerged as a recurring decision heuristic (See Section 5.3).

Our analysis also revealed factors that had small but reliable effects. Whether a county was located on the border of a state or in its interior showed only modest effects as shown in Figure 7. Counties on the state boundary showed slightly higher sensitivity to evidence than interior counties. Design anchored JND was lower for boundary counties in all eight conditions, but only SVSUP maps, decision pixel maps, ordered decision pixel maps, and probability exceedance maps had intervals that excluded zero. Design anchored PSE had the largest shifts in ordered decision pixel and animated decision, where boundary counties were selected at a lower evidence threshold than interior ones.

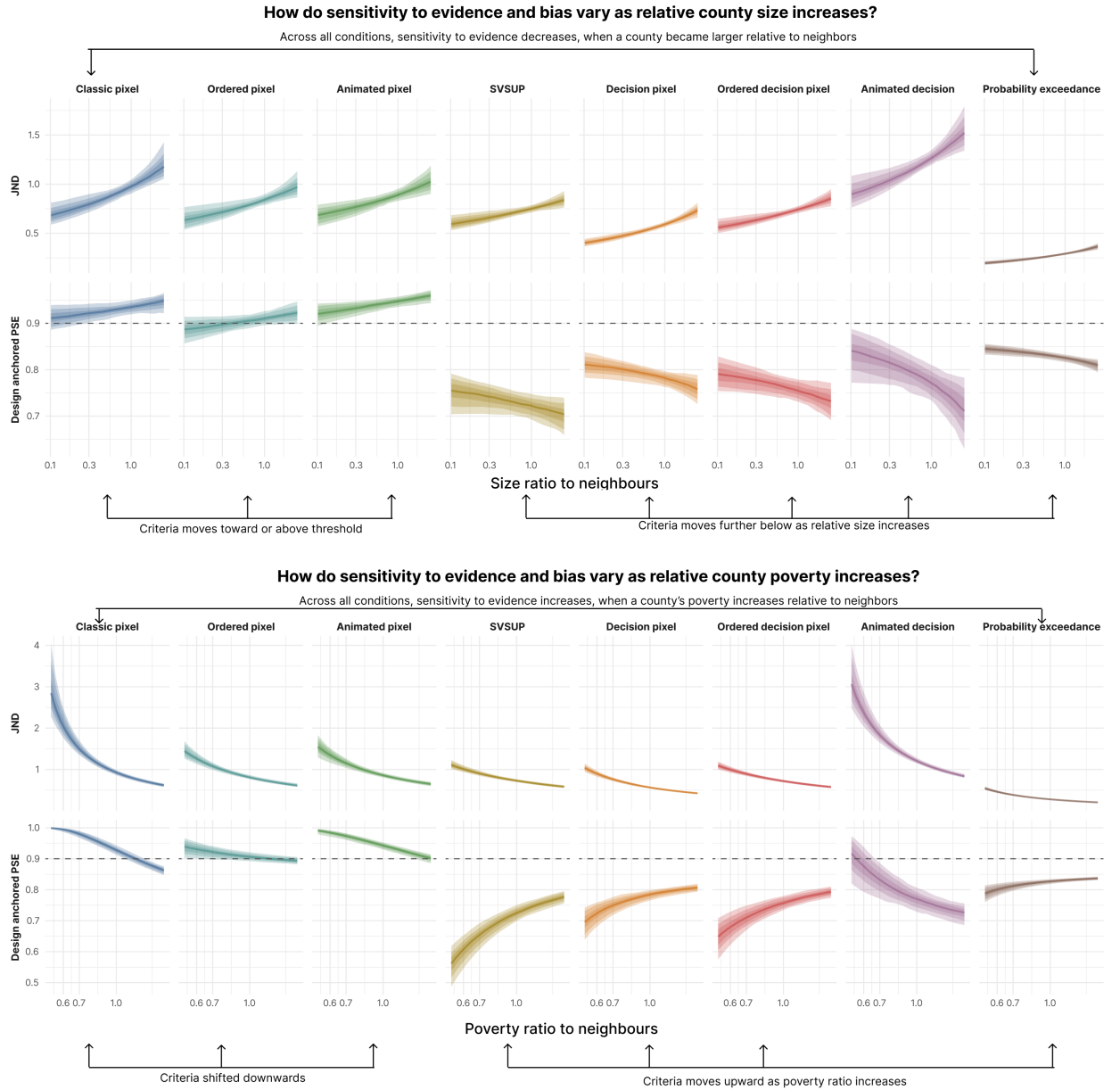


Fig. 6. Effects of county characteristics relative to neighboring counties on sensitivity and decision criteria.

County compactness (Polsby-Popper score) was more consistently associated with participants' decision criterion than with their sensitivity to evidence. Across all measure unit based conditions and probability exceedance maps, the PSE shift was positive and its interval excluded zero. In these conditions, participants selected more irregular counties at lower levels of evidence than more compact counties of the same size. The JND moved less and inconsistently.

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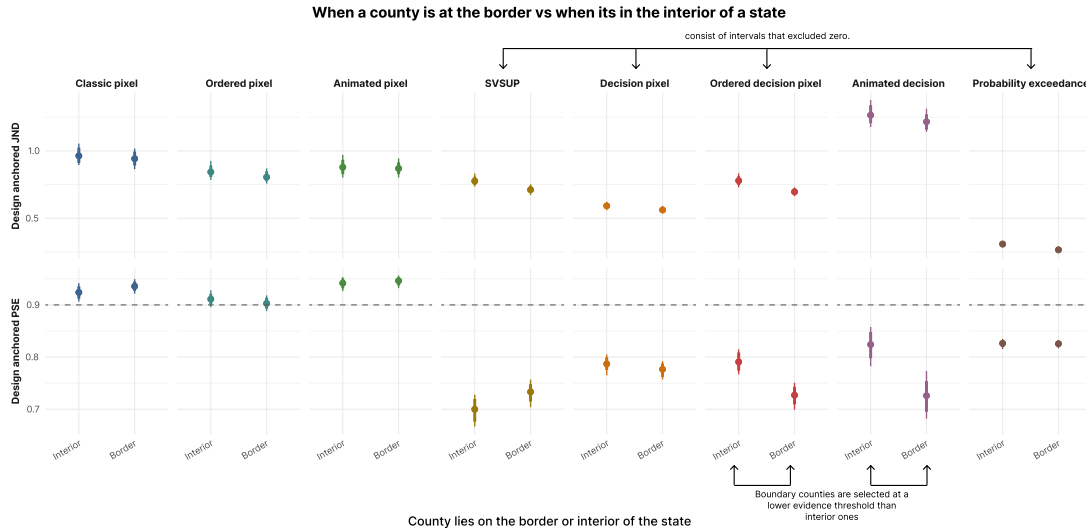


Fig. 7. Effects of county border status on JND and PSE across visualization conditions.

SVSUP was the only condition in which greater compactness was associated with finer sensitivity to evidence. For animated pixel maps, ordered decision pixel maps, and probability exceedance maps, greater compactness was instead associated with reduced sensitivity. We speculate irregular county shapes interfere with visual aggregation or proportion judgments, making it harder to separate the encoded quantity from the geometry of the region. A similar relationship between geographic form and decision-making appears for county area (Figure 5, top).

Some other geographic features we examined, by contrast, showed little systematic association with design anchored JND and design anchored PSE. For instance, Moran's I was reliably associated with design anchored JND in three out of eight conditions (classic pixel, animated pixel, and SVSUP) and the design anchored PSE in one out of eight conditions (animated pixel). One possible reason is that Moran's I varied only by state and task difficulty, providing substantially less within-trial variation than county-level predictors such as relative size and poverty ratio. We retained island indicator as a control but did not include it among other predictors as it was supported by too few counties (less than 0.2%) in our data to estimate per condition.

5.3 Qualitative Analysis

Our qualitative analysis revealed that participants use a small set of recurring strategies to determine their resource allocation decisions. Their responses also revealed several misconceptions about uncertainty and spatial encodings, as well as the use of geographic context and prior knowledge not represented in the data.

Participants relied on a variety of heuristics to interpret the use of color in visualizations. Participants assigned to the pixel and ordered pixel conditions commonly described selecting the *darkest red* counties, effectively judging overall color intensity. In the decision pixel and ordered decision pixel conditions, participants instead looked for counties with the *most red*, judging the proportion of pixels above the decision threshold. Participants in animated conditions similarly attended to which counties remained *consistently red* across frames.

When these judgments were visually complex, participants described additional perceptual strategies for aggregating the color information. Some attempted to reduce the visual complexity of pixel-based maps by perceptually averaging across individual marks. When assigned to the classic pixel maps condition, P182 described their strategy as "I looked for the darkest counties (darker red indicates higher poverty). Sometimes it was obvious, if not I would "soft eye" the map (unfocus the map) so the darker areas stood out a little more. I experimented with turning off the "night light" that makes my screen more orange, but that didn't seem to matter either way." Others considered more deliberate strategies, such as counting individual pixels, but abandoned them when they became too cumbersome. P224 explained, "I tried to see which counties had more red squares than gray squares. If it looked more filled in with red squares, I chose it, because that means it had more areas in the "above threshold poverty rates" category. I thought about trying to count the squares, but there were too many, so I just looked at it and tried my best to pick ones that visually looked more red."

Geographic features also played a role in how participants made their choices. Many participants supplemented color-based judgments with geographic features of the counties themselves. County area was the most commonly mentioned of these features. Some participants used area as a tiebreaker among otherwise similar candidates. For example, P273 said "I looked for the counties that had the most darker red color, if there were more than three, I went with the larger county in the process." In some cases, this strategy highlighted the assumption that geographically larger counties contained more people, revealing a potential conflation of land area with population. Other participants reasoned in the opposite direction and recommended smaller counties because they associated them with rurality or being underserved. As P12 explained, "I tried to use the counties that were smaller and was above the threshold of the requirements because I thought small more rural areas would qualify." This is also a misconception, since many smaller counties tend to be populated urban areas.

Participants also incorporated the spatial context surrounding a county, although neighboring counties played different roles in their reasoning. Some treated clusters of counties in need as corroborating evidence that a county should receive assistance. As P274 described it, "I tried to pick the counties that had as much red as possible and tried to pick those with neighboring counties that were also red." This is inconsistent with a purely perceptual explanation for favoring counties with higher poverty than neighbors due to simultaneous contrast.

Others used neighboring conditions as part of a broader judgment about where resources were most needed. P474 explained, "I looked at the areas that were in need of assistance as a whole, and then I looked at which ones had the least areas not in need around them. I felt that those areas in need surrounded by areas not in need would be more likely to receive support from their more well-off neighbors, and sought to pick areas that ... isolated and therefore potentially more likely to be forgotten about." Still others deliberately chose counties across the state rather than concentrating them geographically. As P557 said, "I tried to find the counties that had the deepest red colors to assist according to the instructions. I then chose counties far apart if possible to diversify the area of assistance."

Participants interwove prior knowledge and fairness considerations into their decision-making. Some participants incorporated prior knowledge about places, while others introduced additional values about who should receive assistance and how resources should be distributed. P475 described a strategy that combined both: "... I looked for Native Reservations, because we owe them so much ... I looked to spread the money throughout areas of the state, thinking that helping three farther flung areas would help a larger population geographically. Sometimes I could not choose, and so I eliminated deep red counties next to the wealthiest counties, because of some thought or rationale that the locality should have more of a role when it is so wealthy. I eliminated Onondaga because I've lived there, ... and there were so many other areas in NYS in need."

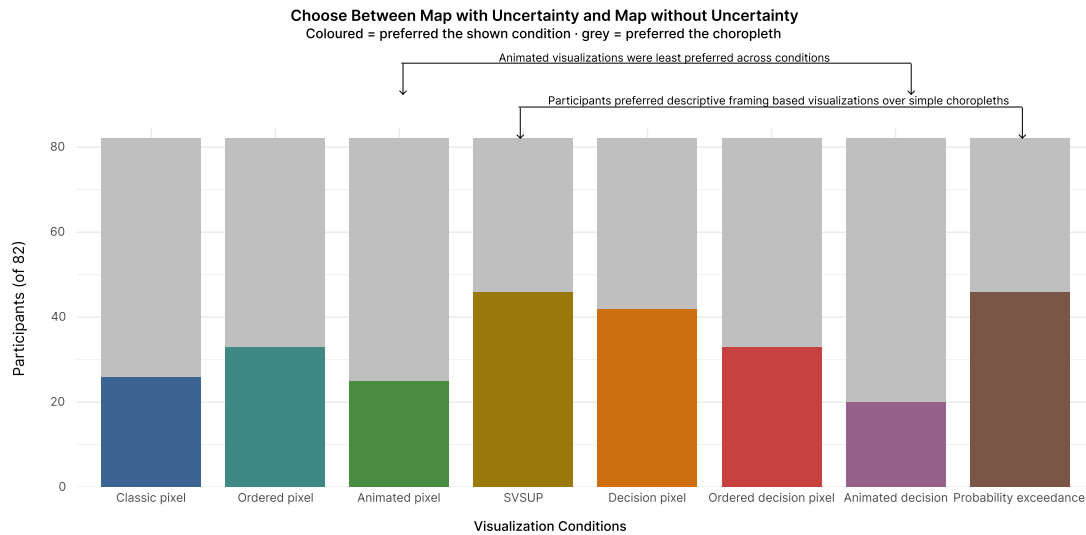


Fig. 8. Participant preference for each uncertainty visualization compared with a conventional choropleth.

Other participants used prior geographic or socioeconomic knowledge to interpret the map. P185 explained, "When I recognized a state, I tried to use what I know about the state, things like geography, areas of industry, large cities and so on, to determine where I think the greatest poverty may be. For example, I know much of the Appalachian region of West Virginia is very poor, so I selected counties that border the mountain range. And, I know Philadelphia is very poor, despite being a major city, so that was easy to pick out. Generally, the more rural a county, my expectation the county was poorer was greater."

Asking participants to take on the role of a consultant also appeared to invite a sense of responsibility that shaped some participants' choices. Instead of identifying the counties that satisfied the decision rule, some participants considered which places seemed underserved or especially deserving of aid. As P80 reflected, "Initially, I wanted to consider their geographic location and size primarily, while also only considering the ones marked with apparent red denoting above threshold poverty rate, with more remote or smaller counties being favored as I feel they are likely underserved. As I went, I simply tried to focus more on the ones that are most red and thus most clearly above the poverty threshold." Responses like these challenge the notion that deviation from a normative decision rule [13], like the 0.90 threshold we instructed, should be straightforwardly interpreted as an inability to use the provided information to make an optimal judgment. Clearly, some participants were contesting the provided decision rule itself as a criterion.

Participants often preferred maps without uncertainty for multiple reasons. When asked whether they preferred a conventional choropleth map without uncertainty information over the uncertainty visualization assigned to them, 385 of 656 participants selected the choropleth (see Figure 8). This preference is consistent with prior visualization research showing that viewers may discount, avoid, or have difficulty reasoning with uncertainty [12]. Participants' explanations suggest that this preference arose for several different reasons.

Most often, participants described choropleths as easier and simpler to read. For example, P452 stated, "Choropleth Map is very clear and easy to interpret thus I would prefer it". For others, the preference reflected difficulty understanding what the uncertainty representation meant. P333 explained, "The map on the right [choropleth] is better for me. As

much as the map on the left tells me the poverty rates along with uncertainty information, it makes me doubt the visual because the uncertainty cannot be explained". In some cases, participants appeared to interpret uncertainty itself as making the underlying data less trustworthy. As P101 stated, "I would feel more certain if it was just the poverty rates only. The other uncertainty information could cloud reality". These responses suggest that explicitly communicating epistemic uncertainty could sometimes reduce participants' confidence in the visualization rather than help them reason about the reliability of the estimates.

Participants also sometimes characterized uncertainty visualizations as noisy, inaccurate, or more susceptible to error. Notably, when comparing SVSUP and choropleth maps, P171 responded, "The Choropleth map is much easier to read with less bias. I would be able to determine facts much more easily with that one."

Other responses revealed deterministic construal errors [33], in which representations of possible outcomes are interpreted as concrete states of the world. In frequency-framing based conditions, some participants appeared to interpret sampled outcomes deterministically. For pixel-based representations, a few participants misunderstood individual pixels to represent geographic subareas within a county. As P409 describes it "I would rather choose the pixel map for decisions like this, so you can see which parts specifically of the county need help the most." Similarly, some participants interpreted changes across frames in animated conditions as changes in the underlying data distribution over time rather than alternative plausible outcomes.

Finally, some participants preferred choropleths because they asserted these maps offered more information than the uncertainty visualization. This was particularly salient for decision-unit representations, which intentionally reduced the data to whether sampled outcomes fell above or below the decision threshold. P385 explained, "The map on the right is significantly more helpful to completing this task because it does not just offer a binary yes or no above the threshold poverty rate but instead shows a continuous scale represented through shaded colors to demonstrate the exact relative poverty levels of each county." Although the decision unit visualizations use entropy-reducing data transformations, these transformations are actually less lossy than showing only a point estimate. We speculate that some participants get this backwards because the choropleth map makes its information loss less visible to the user.

6 Discussion

We present a large crowdsourced experiment examining resource-allocation decisions across eight variations on choropleth maps designed to show uncertainty (Figure 1). We chose these conditions to vary in two design dimensions that are important for uncertainty representation: framing (i.e., frequency vs description) and units (i.e., original measures vs decision-specific summaries). Our findings not only inform on how well these representations support decisions, but also how human interpretation interacts with geospatial variables that are common design constraints when creating visualizations spatial data. We summarize these as high-level takeaways.

No visualization to rule them all. Our findings highlight that there is no single visualization condition that performs best across all measures. Consistent with prior work cautioning against simple rankings of visualization effectiveness [27], we instead find a sharp tradeoff between sensitivity to evidence and bias relative to the normative criterion. We find that visualizations representing uncertainty as decision units (e.g., exceedance probabilities and other conditions derived from them) and SVSUPS demonstrate what we call *the canonical effect of entropy-reducing data transformations*: by offering statistical summaries that may oversimplify the underlying data distribution, they reduce variance in decision-making while inducing a bias to allocate resources too readily compared to the instructed criterion. On the other hand, visualizations that frame uncertainty as frequency in measure units (e.g., classic pixel maps, ordered pixel maps, and animated pixel maps) show intermediate sensitivity but support decision-making at

or above the criterion despite showing more information than the task demands. We thereby encourage readers to approach the question of which visualization to use in a given application as a balancing act, one that depends on geographic context.

County area plays an important role in how people make decisions. Our quantitative as well as qualitative analyses highlight the importance of geographic features in decision making. Across all geospatial predictors and conditions, the area of a county is the only one where, as the predictor (area) increases, participants become less sensitive to evidence and more biased across every condition. Participants also echoed this pattern in their survey responses. This suggests that viewers may treat geographic extent as meaningful even when it is irrelevant to the decision rule, sometimes using it as a proxy for population, rurality, or need [34]. Designers of geospatial decision aids should therefore consider whether map area introduces unintended cues and, when necessary, supplement maps with decision-relevant contextual information such as population.

The number of counties impacts different visualization conditions in different ways. Similar to the effect of county size above, our quantitative results show strong effects of number of counties on decision quality, above and beyond the mere effect of added distractors, which is accounted for by the measures described in Section 4.5. However, the effect for number of counties varies across visualization conditions. For VSUPS and visualizations with frequency framed decision units, increasing the number of counties exerts a strong bias to allocate resources with too little evidence of need. For animated pixel maps (a.k.a., HOPS [14] applied to choropleths), we see a steep loss in sensitivity and overly conservative decision-making as the number of areal units increase. This suggests that HOPS scale poorly to large numbers of distributions, which makes sense given that human perception struggles to track many moving objects simultaneously [37]. Classic pixel and ordered pixel maps appear most robust to these effects.

I am my neighbors' neighbor. We found that county selection is also associated with the relative size and poverty rate of a county compared to its neighbors. Across all conditions, we see that sensitivity to evidence declines as a county becomes larger than its neighbors. Decision criteria also became more similar across most conditions as relative poverty increased, converging toward the instructed decision rule. Participants likewise described using neighboring counties both to support and to rule out potential selections. This suggests that local comparisons are an important feature of geospatial decision-making. Designers should therefore preserve the ability to compare neighboring areas. Binary encoding, like that offered by decision-unit maps, may make these comparisons more difficult by suppressing differences within counties.

6.1 Limitations

Most limitations of our study stem from our choice of task, which asked participants to select three counties in a given state map to receive aid based on apparent need. This choice constrains the process that we use to generate our stimuli. Specifically, we use real ACS data as a seed for the data we show participants but then modify it to fit our study design (see Section 4.3). Key decisions that depart from the original ACS data include fixing the grand mean poverty across states and adjusting the rank-ordered county-level estimates to control task difficulty and the number of “correct” answers per trial. Although this allowed us to systematically manipulate visualized evidence while preserving characteristics of real U.S. county data for poverty rates, naturally occurring datasets may contain statistical and spatial relationships not represented in our experiment. For instance, real data show greater heterogeneity in measures like poverty, and such quantities may be naturally correlated with geospatial factors in different ways than they were in our experiment. Nonetheless, our study design makes an earnest attempt to “let maps be maps” while still achieving balance across key predictors, rather than trying to synthetically control every aspect of presentation.

Additionally, our choice of task may not be representative of every scenario where geospatial uncertainty visualizations are used. Our study specifically examines how people reason about the tails of distributions in a resource-allocation task, and its findings may not reflect the broader utility of these visualizations for geospatial exploration or other tasks that aren't about the cumulative probability of extreme events. Although our choice to adopt a decision task with a normative decision rule [13] enables us to rigorously quantify bias for this class of decision, it does not yield a globally applicable measure of response bias. As such, our results about sensitivity to evidence should be understood to apply more broadly to the perceptions of tail probabilities, whereas our results about bias should be interpreted as representing behavior under an explicit incentive regime. However, insofar as participants' qualitative responses suggest that they incorporate personal knowledge and values to think beyond the instructed decision rule, our PSE results may be better conceptualized as descriptive of the relative tendency to allocate resources across conditions of our experiment.

Finally, our crowdsourced participant population may reason about decision-making using these visualizations differently from experts in spatial data science. Experts have more context about problems in their domain, so they likely bring different personal knowledge and values to the interpretation of spatial data. Our results suggest that such factors weighed in our participants' reasoning, so we expect results may vary across levels of expertise, and also the subject matter and decision or policy at hand. Our study design can be replicated, extended, or modified in these other settings to provide rich evidence about decision quality and how it interacts with geospatial variables.

7 Conclusion

We contribute findings from a pre-registered, crowdsourced mixed-methods study to investigate how different visualization conditions influence geospatial decision-making under a resource allocation task. We examine eight conditions — classic pixel, ordered pixel, animated pixel, shrinkage value-suppressing uncertainty palette, decision pixel, ordered decision pixel, animated decision, and exceedance probability maps. Our study measures decision quality with these representations across a variety of realistic geospatial conditions. We aim this experiment and the exposition of its findings toward informing displays in spatial data science software and decision aids in a policy setting.

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